

Lesson Plan

Introduction to Biological Anthropology | 4 Class Periods | Computer-Based Investigation

Course Context

Students are told that a fossilized primate specimen has been recovered and must determine how it moved and how large it was, using only skeletal measurements and a comparative dataset of 478 modern primate specimens. The investigation is built so that no single line of evidence resolves the question. Students must integrate four independent analyses and construct a written argument that explicitly acknowledges uncertainty, rather than pattern-matching their way to a single “correct” answer.

Level	Format	Duration	Grouping
Introductory undergraduate biological anthropology	In-class, instructor-led, browser-based (Pyodide/Python in-browser, no login or install)	4 class periods, ~50–75 min each	Individual or pairs; assign different mystery specimens across groups

Learning Objectives

By the end of this investigation, students will be able to:

- Explain how limb proportions, body mass, olecranon morphology, and cross-sectional bone geometry each provide independent, partial evidence about primate locomotor behavior.
- Calculate and interpret the intermembral index (IMI) and olecranon-ulna index (OUI) from raw skeletal measurements.
- Estimate body mass from femoral head diameter using taxon-specific regression equations, and evaluate whether a given estimate is biologically plausible.
- Analyze cross-sectional geometric properties (cortical area, $I_{\max}/I_{\min}$, polar moment J) as evidence of habitual mechanical loading, consistent with Wolff's Law.
- Construct a written scientific argument that states a claim, cites specific supporting evidence, explicitly acknowledges conflicting evidence, and reasons about adaptive significance, under conditions of genuine uncertainty.
- Evaluate the limits of single-metric inference in comparative morphology and articulate why converging, independent lines of evidence strengthen a scientific claim.

Alignment with Vision and Change Core Competencies

This investigation is not tied to K-12 NGSS standards, since it targets an undergraduate course. It is instead framed against the AAAS *Vision and Change in Undergraduate Biology Education* (2011) core competencies, a widely referenced framework in undergraduate life-science curricula. Instructors should adapt this mapping to their own institution's stated learning outcomes as needed.

Core Competency	How this investigation addresses it
Ability to apply the process of science	Students generate and test a locomotor hypothesis against four independent, quantitative lines of skeletal evidence.
Ability to use quantitative reasoning	Students calculate IML, OUI, and regression-based body mass estimates, and interpret scatter plots, box plots, and log-scale distributions.
Ability to use modeling and simulation	Interactive scatter/box plots and regression outputs are generated live in-browser from a 478-specimen comparative dataset.
Ability to communicate and collaborate	Students construct a structured written argument and participate in a whole-class discussion synthesizing group findings across different mystery specimens.
Understanding the relationship between science and society	The reveal-and-discuss structure models how paleontological and forensic inference actually proceeds under incomplete evidence and genuine uncertainty.

Materials & Prerequisites

- Computer with a modern browser (Chrome, Firefox, or Edge) and internet access.
- Free Morphosource.org account, created before Day 1.
- epiSTEMic platform: instructor.html, student-landing.html, and investigation.html — no login or installation required.
- Browser-based ImageJ (ij.imjoy.io) and the MomentMacro_CSATS.txt macro file, needed for Day 3.
- One mystery specimen assignment per student or group (MS-11, MS-12, MS-13, or MS-14); see the instructor reference table in instructor.html for identities and pedagogical tensions. Do not share specimen identities with students before the Day 4 reveal.
- Slide decks: slides-limb-proportions.pptx.pdf, slides-biomechanics.pptx.pdf, slides-csg.pptx.pdf, slides-post-discussion.pptx.pdf.

Pacing Overview

Day	Focus	Platform	Slide deck(s)
1	Limb Proportions & Locomotion	Tab 1	slides-limb-proportions.pptx.pdf
2	Body Mass & Biomechanics	Tabs 2–3	slides-biomechanics.pptx.pdf, slides-post-discussion.pptx.pdf
3	Cross-Sectional Geometry	Tab 4	slides-csg.pptx.pdf
4	Argument Construction & Reveal	Tab 5	none — discussion-based

DAY 1 Limb Proportions & Locomotion

Platform: Tab 1 — Limb Proportions | Slide deck: slides-limb-proportions.pptx.pdf

Today's objectives

- Define the intermembral index (IMI) and explain what it indicates about locomotor behavior.
- Measure four limb bones (humerus, radius, femur, tibia) for an assigned specimen in Morphosource.
- Plot a specimen against the 478-specimen comparative dataset and interpret where it clusters.

Suggested timing (75-minute period; trim discussion time for 50-minute periods)

Time	Segment
10 min	Hook: show 1–2 unlabeled locomotion videos. Students describe what they see and brainstorm what skeletal features might explain the differences.
20 min	Direct instruction (slides-limb-proportions.pptx.pdf): tetrapod origins, locomotor diversity, bones of interest, the IMI formula, measurement protocol.
10 min	Class models a measurement together in Morphosource using a non-mystery example specimen.
25 min	Students measure their assigned mystery specimen's four limb bones in Morphosource, enter values into investigation.html Tab 1, and explore the scatter plot and IMI box plot.
10 min	Turn-and-talk: "Where does your specimen cluster? What does that suggest — and what doesn't it tell you yet?"
FORMATIVE CHECK Cold-call 2–3 groups to state their specimen's IMI and one locomotor category it is consistent with, plus one category it rules out.	

PREP FOR DAY 2

No assignment required. Optionally, ask students to jot down what additional evidence (beyond limb proportions) would help narrow their locomotor hypothesis.

DAY 2 Body Mass & Biomechanics

Platform: Tabs 2–3 — Body Mass, Olecranon Index | **Slide deck:** slides-biomechanics.pptx.pdf, slides-post-discussion.pptx.pdf

Today's objectives

- Estimate body mass from femoral head SI diameter using taxon-specific regression equations, and evaluate which estimate is biologically plausible.
- Explain the elbow as a third-class lever and the olecranon process as the triceps insertion point.
- Calculate the olecranon-ulna index (OUI) and interpret it against six locomotor group ranges (Drapeau, 2004).

Suggested timing

Time	Segment
10 min	Warm-up: quick-share Day 1 findings. Pose the question: “Does knowing body size change your locomotor hypothesis?”
20 min	Direct instruction (slides-biomechanics.pptx.pdf): levers and mechanical advantage, the elbow as a third-class lever, the olecranon process.
15 min	Students measure femoral head diameter in Morphosource, enter it into Tab 2, and compare the five taxon-specific mass estimates — discussing which is plausible for their specimen and why the others are not.
15 min	Students measure olecranon length and ulnar diaphyseal length, enter values into Tab 3, and plot OUI against the six locomotor group ranges.
10 min	Closing mini-lecture (slides-post-discussion.pptx.pdf): ulnar morphology across locomotor groups, comparative literature data.
5 min	Exit ticket: “What does your OUI rule in or out? Where does it overlap with other locomotor groups?”

FORMATIVE CHECK

Collect exit ticket responses; scan for students conflating “overlap” with “no information” — overlap still narrows the field even when it doesn't resolve it.

PREP FOR DAY 3

Confirm each student/group has access to (or knows how to access) their assigned specimen's CT cross-section images ahead of the Day 3 ImageJ workflow.

DAY 3 Cross-Sectional Geometry

Platform: Tab 4 — Bone Geometry | Slide deck: slides-csg.pptx.pdf

Today's objectives

- Explain Wolff's Law: bone as living tissue that structurally responds to habitual mechanical loading.
- Run the ImageJ MomentMacro on CT cross-sections to extract cortical area, $I_{\max}$, $I_{\min}$, and polar moment J.
- Interpret cortical area as a robusticity proxy and $I_{\max}/I_{\min}$ as a proxy for loading directionality.

Suggested timing

Time	Segment
10 min	Warm-up: a brief analogy for Wolff's Law (e.g., bone density differences in a tennis player's dominant vs. non-dominant arm).
15 min	Direct instruction (slides-csg.pptx.pdf): bone as living tissue, Wolff's Law, cortical area, $I_{\max}/I_{\min}$ as a loading-directionality proxy.
10 min	Class demo of the ImageJ + MomentMacro workflow on a sample image.
25 min	Students run MomentMacro on their specimen's femur/humerus CT images, upload the resulting log files to Tab 4, and explore the J scatter plot, $I_{\max}/I_{\min}$ box plots, and cortical area box plots.
15 min	Small-group discussion: "Does your bone geometry data support or complicate the locomotor hypothesis you had after Days 1–2?"
FORMATIVE CHECK Each group states, in one sentence, how their bone geometry data either strengthens or complicates their working hypothesis.	

PREP FOR DAY 4

Students should arrive at Day 4 with Tabs 1–4 completed in the platform — those results auto-populate the summary pills on Tab 5.

DAY 4 Argument Construction & Reveal

Platform: Tab 5 — Build Argument | Slide deck: none — discussion-based

Today's objectives

- Synthesize four independent lines of evidence into a structured written scientific argument.
- Explain why acknowledging conflicting evidence strengthens, rather than weakens, a scientific claim.
- Evaluate the limits of the available evidence after the specimen identity is revealed.

Suggested timing

Time	Segment
10 min	Students revisit their auto-populated summary pills (Tabs 1–4) and briefly discuss their overall picture with a partner.
5 min	Direct instruction: what makes a scientific claim defensible; why conflicting evidence belongs in a strong argument, not just a weak one.
20 min	Students individually write their four-part argument in Tab 5 (claim, supporting evidence, conflicting evidence, adaptation/natural selection reasoning) and submit for AI formative feedback.
10 min	Students revise their argument based on feedback, then unlock the reveal.
20 min	Whole-class discussion (see prompts below).
5–10 min	Wrap-up: connect back to the Vision and Change competencies; preview how this mirrors real paleontological and forensic inference under uncertainty.

Discussion prompts

- Which metric was most useful? Which was most misleading?
- What additional evidence would have helped?
- How does this relate to how paleontologists actually work?
- What would change if the specimen were from a different habitat?

NOTE

The reveal is deliberately gated behind argument submission so students cannot skip to the answer. The follow-up question — “*What does this tell you about the limits of your evidence?*” — is where the deepest learning happens; do not skip the post-reveal discussion for time.

Suggested Assessment Rubric

A simple 4-point holistic rubric for the Day 4 written argument. Adapt point values to your course's grading scheme.

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Claim clarity	Claim is specific, testable, and appropriately hedged	Claim is clear but under- or over-stated	Claim is vague or only partially supported by the analysis	No clear claim, or claim contradicts the student's own data
Supporting evidence	Cites specific values from 3–4 analyses with correct interpretation	Cites specific values from 2–3 analyses	Evidence is vague or drawn from only one analysis	No specific quantitative evidence cited
Conflicting evidence	Explicitly names conflicting data and explains its significance	Names conflicting data with limited explanation	Vague acknowledgment that uncertainty exists	No conflicting evidence acknowledged
Adaptive reasoning	Connects morphology to function to natural selection with correct mechanism	Connects morphology to function, reasoning less developed	Restates data without functional or evolutionary reasoning	No adaptive reasoning present

Differentiation & Notes for Instructors

- Assign different specimens (MS-11 through MS-14) to different groups to create productive disagreement in the Day 4 discussion — see the instructor reference table in [instructor.html](#) for identities and pedagogical tensions. Do not share this table with students before the reveal.
- For 50-minute periods, split Day 2's content across two sessions, or assign the post-discussion slide deck as a short reading instead of an in-class lecture.
- For partially remote or asynchronous sections, Tabs 1–4 can be completed as homework, reserving synchronous class time for the Day 4 discussion and reveal — that is where the deepest learning occurs.
- No backend, login, or installation is required for the epiSTEMic platform itself. Only the free Morphosource account and browser-based ImageJ require separate setup, and only ImageJ is needed just for Day 3.

References

Drapeau, M. S. M. (2004). Functional anatomy of the olecranon process in hominoids and Plio-Pleistocene hominins. Source for the six locomotor group OUI ranges used in Tab 3.

American Association for the Advancement of Science (AAAS). (2011). *Vision and Change in Undergraduate Biology Education: A Call to Action*. Source for the core competency framework referenced above.